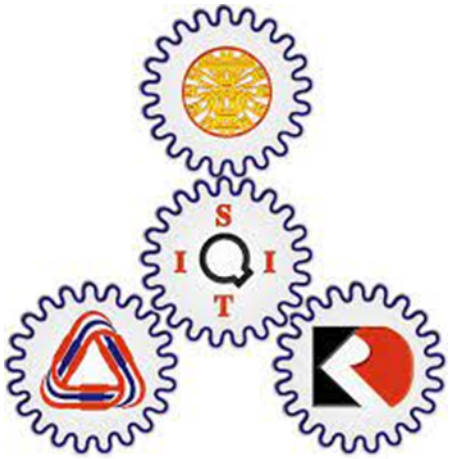
Sirindhorn International Institute of Technology



Semester 2/2023

ITS/DES229 Human-Computer Interface Design

Group Assignment #4 - Usability Testing

Group 8: *Vraiment Cool*

Project Topic: Play N' Learn

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Objectives

• Conducting a usability test using the Think-Aloud Protocol for the prototype of our product, in order to create a list of defects for further revisions to the prototype.

Introduction

This document will serve as a report file for ITS/DES229 2023-2 Group Assignment #4. In this assignment, we will be conducting a usability test using the Think-Aloud Protocol for our product. To begin, a set of goals & criteria are created to outline what we wish to accomplish with this test. Then, participants are gathered to assess the prototype based on the outlined goals & criteria. Lastly, responses from the participants are recorded and analyzed to form a set of defects with our prototype. This will help us to refine the prototype further as we continue to develop the project.

Project: Play N' Learn

Our product is an app to help users create a "game-ified" learning experience. Borrowing concepts from flashcard-type learning games (e.g. World Geography, Duolingo), we intend to provide a platform where users can create similar experiences with any subject they'd like. Users can either choose to create an experience (for educators) or choose to play it (for students) - with a built-in progression system similar to games. (e.g. leveling up, progression paths)

Plan of Action

Goals

- Testing ease of access to various functions in our product. (e.g. Authentication, Message Inbox, Classroom Creation)
- Testing whether student users will pay more attention to lessons when presented in a gamified format.
- Testing whether the app is able to promote self-directed learning in students.
- Testing whether teacher users are able to host a classroom using our app.

Participants

Participants are divided into two groups: students and teachers. For our usability evaluation, we have gathered 3 students and 1 teacher throughout the test period.

- **Ms. Mutita Phansamdang** is a Year 2 student representative of SIIT.
- **Mr. Bhone Khant Kyaw** is a Year 2 student representative of SIIT.
- **Mr. Puwarit Khowean** is a Year 2 student representative of SIIT.
- **Mrs. Sakunee Srihera** is a teacher representative of Kannasoot Suksalai School, Foreign Languages Department.

Tasks/Criteria

Each participant is invited to an online personal session lasting about 15 to 30 minutes via Zoom. The participants are instructed to evaluate the UI images of our prototype based on the following tasks/criteria, and any reactions and impressions are noted down for analysis.

Tasks for students:

- Log in and see the enrolled classroom.
- Review the explanation provided by the teacher after a quiz.
- Tap on a classroom to see options like materials and classwork.
- Completing a quiz and learning from mistakes.
- Testing offline functionality (Optional)

Tasks for teachers:

- Setting up a new classroom.
- Building a course outline
- Creating engaging activities with interactive exercises.
- Testing live pop quizzes

Analysis & Summary

Defects

The defects discovered through the test in our prototype design can be summarized as follows.

1. ***Difficulty for mobile devices:***

   Some participants commented that the UI seems difficult to use/cluttered on mobile devices.

2. ***Not enough analytics info:***

   The teacher participant commented on how the classroom analytics page includes too little information to be usable.

3. ***Ill-defined methods to create & interact with the interactive exercises:***

   Most participants struggle to imagine interactions with the interactive exercises portion of our prototype, suggesting that this part needs more work to fully flesh out.

Summary & Improvement Recommendations

Firstly, the key point of defect found from the test is the menu layout for mobile devices. Most participants reported feeling that the UI feels somewhat too cluttered for mobile displays. Methods to rectify this issue may include utilizing option groupings to organize the layout better.

Additionally, the teacher participant gave complaints about the classroom analytic page not having enough information. Improvements can be made by extending the page as a slider, in order to be able to fit more information; such as the exercise completion status of each student.

Lastly, there are complaints from both sides, saying that the interactive exercises in our prototype need more work to fully flesh out the functionality - at this stage, the participants could not imagine navigating this function of our prototype.